



Vietnam National University of HCMC
International University
School of Computer Science and Engineering



Object Oriented Programming

Design Patterns

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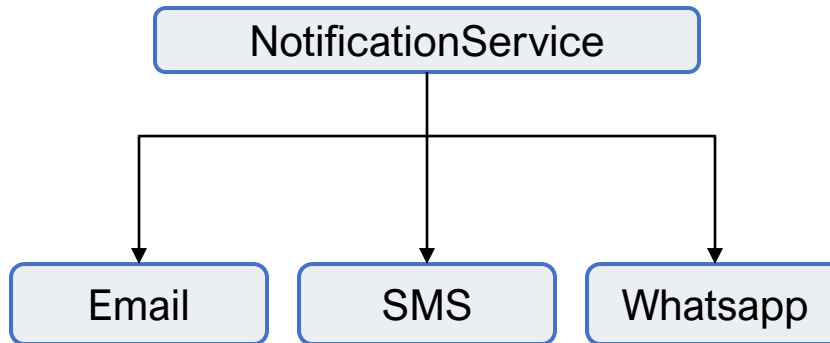
Lecture Objectives

- ☑ Understand the Concept of Design Patterns
- ☑ Recognize the significance of Design Patterns in Software Development
- ☑ Apply Design Patterns in real world case study

What are Design Patterns

Design patterns are typical solutions to commonly occurring problems in software design. Design patterns in software development provide established solutions to solve recurring design problems.

Example: A notification system supports sending messages by variants of apps.



Why Design Patterns?

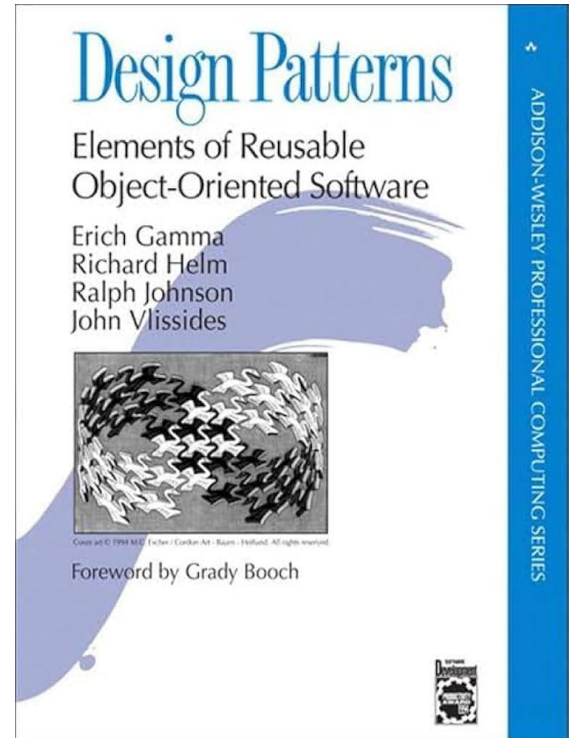
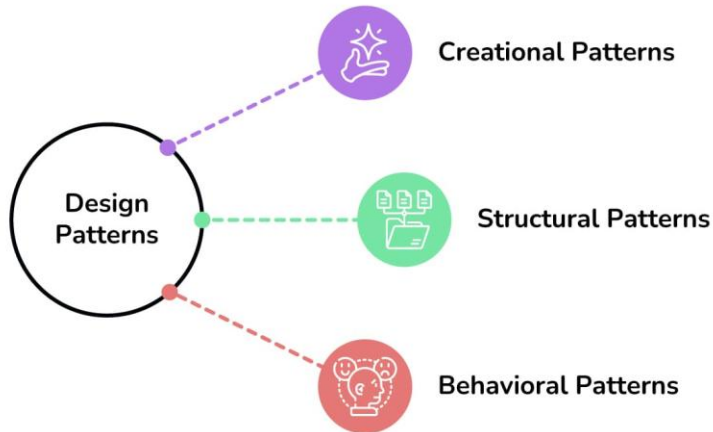


Offshore
Outsourcing



Hiring In-House
Teams

Gang of Four (GoF) Design Patterns



History and Origins of Design Patterns

1**Christopher Alexander (1977)**

"A Pattern Language" introduced the idea of architectural patterns.

2**Gang of Four (1994)**

The book "Design Patterns: Elements of Reusable Object-Oriented Software" introduced 23 design patterns.

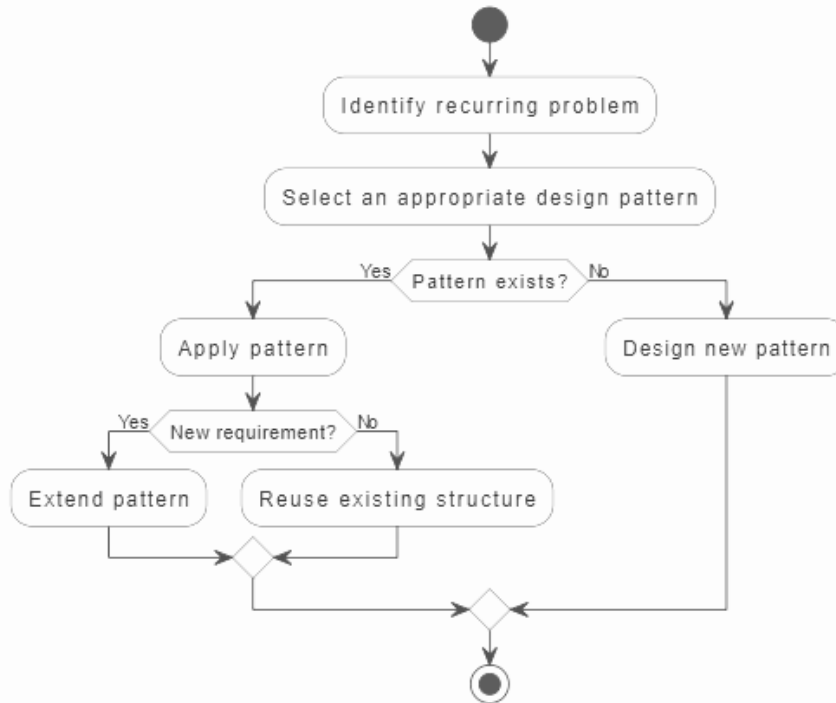
3**Further Evolution**

The field expanded into various domains, including enterprise architecture, UI design, and distributed systems.

Why Design Patterns Matter in Development

- ☾ Simplifies Communication: Patterns give developers a shared language to discuss design approaches.
- ☾ Improves Code Readability and Maintainability: Patterns provide a standardized way to solve problems.
- ☾ Enhances Reusability and Scalability: By reusing solutions, development becomes more efficient.

How to construct a Design Pattern?

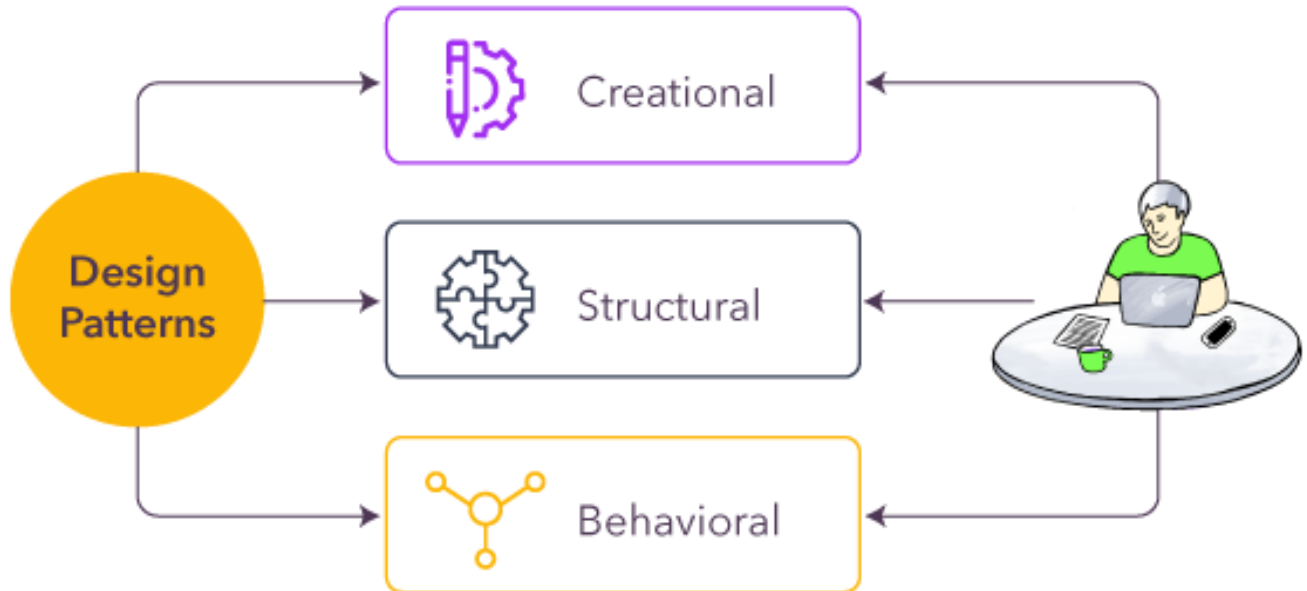


Structure of a Design Pattern

Four essential components of a design pattern:

- **Name:** A concise identifier for easy reference (e.g., Singleton, Observer).
- **Problem:** The specific issue or recurring situation the pattern addresses.
- **Solution:** The approach or structure that resolves the problem.
- **Consequences:** The pros and cons of using this pattern.

Patterns



Source: <https://www.jobsity.com/blog/design-pattern-software>

Creational Patterns

Define Creational patterns as solutions focused on *efficient object creation*.

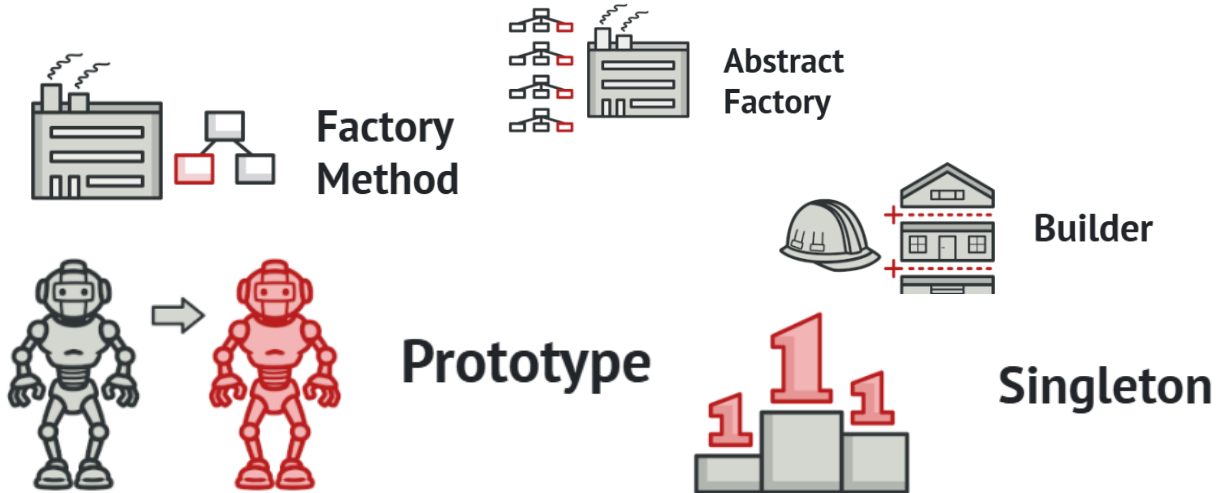
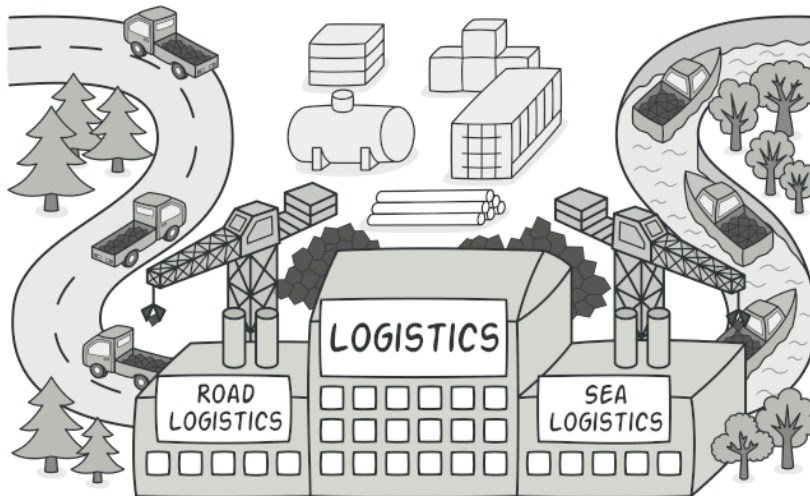


Image credit: <https://refactoring.guru>

Factory Method Pattern

Factory Method: a *creational design pattern* that provides a way to create objects without specifying the exact class of the object being created.



Factory Method Pattern

Components:

- **Product Interface:** Defines the type of object to be created.
- **Concrete Product Classes:** Implement the Product interface for specific types.
- **Creator (Factory) Class:** Contains the factory method that creates and returns objects
- **Concrete Creators:** Subclasses of the factory class that implement or override the factory method to produce specific product types.

Factory Pattern

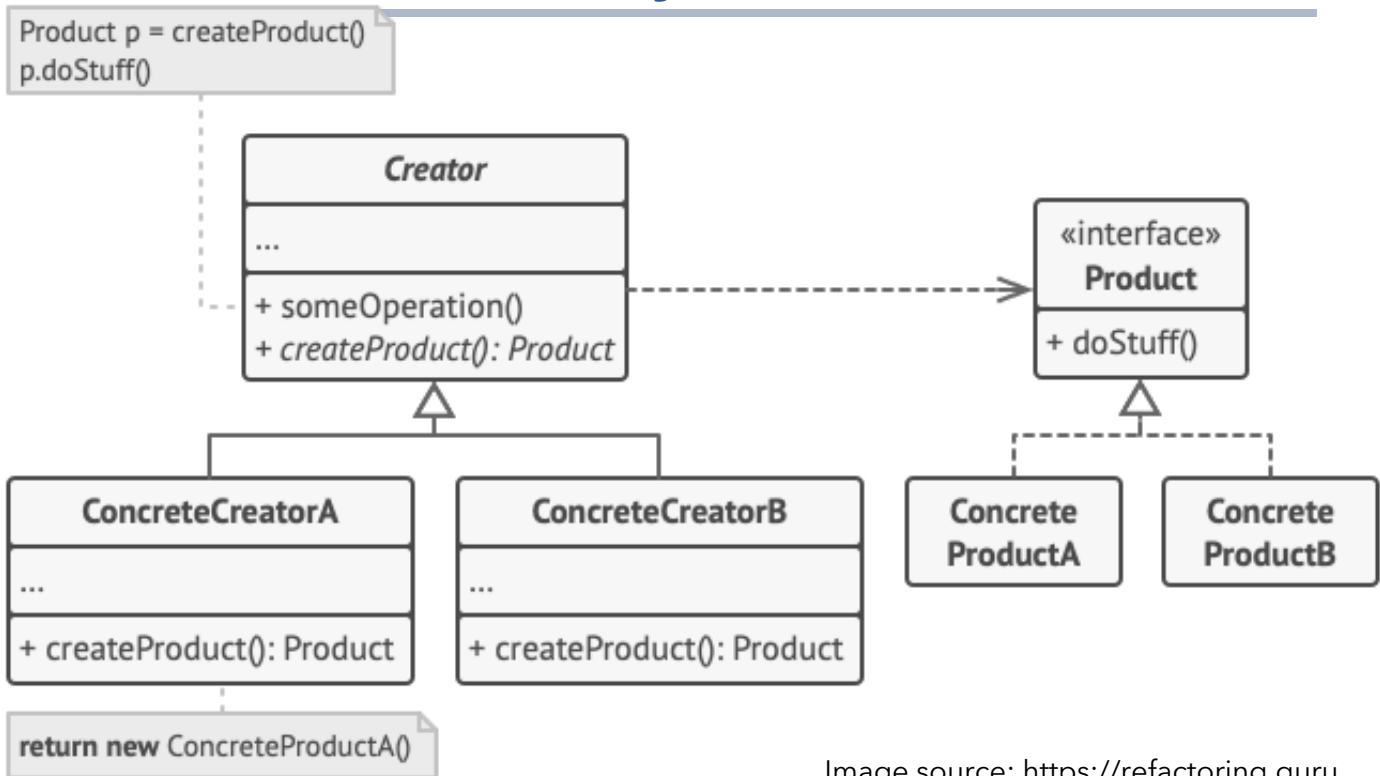


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